

Davian R. Chin

Scientific Machine Learning Researcher — Fractional Calculus & Neural Dynamics

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Research Profile

Scientific machine learning researcher and computational scientist (PhD candidate; Fellow of the IMA) working at the intersection of fractional calculus, stochastic processes, operator theory and deep learning.

My PhD studies trustworthy long-range dependence (LRD) inference in finite stochastic time series through audited benchmarking, conditional theory (the *Variance Trap*) and a proposed null-relative dependence field. It separates fine-scale path regularity, tails and nonstationarity from coarse-scale memory.

I build reproducible research software in Python (JAX/PyTorch). Brain-inspired models motivate the work; clinical validation is a longer-term application. Seeking research-scientist roles in scientific ML.

Research Interests

- Long-range dependence and heavy-tailed stochastic processes; L^p /Besov path regularity and wavelet characterisations of roughness.
- Fractional calculus, fractional-order differential equations, functional analysis and operator theory.
- Scientific ML: neural operators, neural (fractional) ODEs/SDEs, physics-informed learning, fractional reservoir computing.
- Criticality, scale-free dynamics and self-organisation in brain networks; biomarkers of neurodegeneration.

Education

PhD, Biomedical Engineering (Computational Neuroscience) University of Reading, UK – 2025–present (expected 2028)

Programme: *Trustworthy inference of memory and scale dependence in stochastic dynamics* (supervisors: Prof. W. Holderbaum, Prof. S. Nasuto). Focus: benchmarking, conditional inference limits, null-relative dependence and brain-inspired models.

MSc, Computer Science (AI, ML & Big Data Technologies) University of East London, UK – 2023
Focus: physics-informed neural operators for heat/diffusion PDEs; machine vision; big-data analytics.

MA, Education (Development, Management & Technology) University of South Wales, UK – 2018
Dissertation: computational thinking as a framework for mathematics education.

BSc, Mathematics & Computer Science University of the West Indies, Jamaica – 2008
Pure and applied mathematics with mathematical modelling and operations research.

Research Experience

PhD Researcher — Computational Neuroscience & Scientific ML University of Reading – 2025–present

- **Empirical benchmarking:** created and released `1rdbench`; completed and audited a controlled simulation study of 21 point-estimation pipelines from 12 base algorithms, using 66,000 independent clean records and 161,000 contaminated descendants. Separated point accuracy, interval availability and coverage in the analysis.
- **Conditional theory:** developing the *Variance Trap* account of four distinct limitations: the absence of a finite second-order target under infinite variance, quadratic measurement vulnerability, restricted-observation non-identifiability, and miscalibration of specified estimators. Consolidating exact assumptions and checked proofs.
- **Dependence framework:** proposing a null-relative, functional-indexed coarse-scale field in a multiscale atlas of dependence documents. A restricted study will test classical reduction and a bounded or lower-order extension; fine Sobolev–Besov regularity remains separate from persistent dependence.
- **Dissemination:** released research software with Zenodo DOIs, CI and documentation; invited talk and poster at the School of Biological Sciences ECR Symposium (University of Reading, July 2026). Preparing manuscripts on the corrected benchmark, conditional theory and dependence field.

- Built a physics-informed Fourier neural operator solving the 1D/2D heat equation, studying physics-loss weighting versus Fourier-analysis techniques for data-efficient PDE surrogate modelling.
- Built, trained and deployed CNN/DNN machine-vision models and tree/ensemble/SVM classifiers on a big-data stack; benchmarked optimisation strategies for high-accuracy image classification.

Open-Source Research Software

- **lrdbench** Python – PyPI v1.2.1 – Zenodo 10.5281/zenodo.20937726 – docs at lrdbench.readthedocs.io
Reproducible benchmark framework for LRD estimators: ground-truth, stress-test and observational modes; bundled temporal, spectral, geometric, wavelet, aggregation and data-driven estimator families; contamination operators (heavy-tailed noise, level shifts, outliers, trends); uncertainty-aware interval-coverage reporting; manifest-driven, provenance-complete execution.
- **hpfracc** Python/JAX – Zenodo 10.5281/zenodo.20942679
Pre-alpha research-support library for fractional calculus and fractional dynamical systems in differentiable scientific computing: RL/Caputo/GL operator families, explicit/implicit, non-uniform and adaptive Caputo FDE solvers, experimental neural FODE workflows, probabilistic calibration and stochastic simulation helpers.
- **fourier_pino_model** Python/PyTorch
Physics-informed Fourier neural-operator surrogate for the 1D/2D heat equation.

Publications, Manuscripts & Conference Contributions

- Batic, D., Chin, D. & Nowakowski, M. (2011). The repulsive nature of naked singularities from the point of view of Quantum Mechanics. *European Physical Journal C*, 71, 1624. doi:10.1140/epjc/s10052-011-1624-3
- *Manuscripts in development*: corrected LRD estimator benchmark; conditional *Variance Trap* analysis; restricted null-relative dependence-field methods.
- Talk & poster: *Trustworthy methods for scientific discovery in critical brain research*, School of Biological Sciences ECR Symposium, University of Reading, July 2026.
- Poster: *Fat-tail phenomena and long-memory processes in biological systems*, School of Biological Sciences & Biomedical Engineering Research Conference, University of Reading, 2025.

Technical Skills

- **Scientific computing**: JAX, Equinox, NumPyro, PyTorch, NumPy/SciPy, scikit-learn, pandas, Matplotlib; GPU/CUDA, HPC, Colab/Cloud, experiment tracking.
- **Mathematics & modelling**: fractional calculus and FDEs, stochastic processes and SDEs, functional analysis/operator theory, wavelets, numerical methods, dynamical systems/criticality, Bayesian inference, time-series analysis.
- **Machine learning**: neural operators, PINNs, neural (F)ODEs/SDEs, reservoir computing, surrogate modelling, deep learning (CNN/RNN/LSTM), classical ML.
- **Software & MLOps**: Python (primary), R, MATLAB, C/C++, SQL, Haskell, $\text{\LaTeX}$; Git/GitHub, CI, Docker, Linux; reproducible, provenance-complete workflows.
- **Domains**: computational neuroscience, biomedical engineering, EEG/neural time-series analysis, scientific software engineering.

Professional Memberships

Fellow, Institute of Mathematics and its Applications (FIMA).

References available on request.